

Completed $4N_y$ hard-wall dynamical-critical analysis

Numerical working note
(Dated: September 17, 2026)

We analyze 700 independent Born trajectories of the $N_x = 20$ hard-wall adaptive Gaussian circuit at $N_y = 20, 24, 30, 36, 44, 56, 60$ and depth $T = 4N_y$. All 140 five-sample shards pass checksum, identity, coverage, and numerical validation. The saved occupation spectra reproduce the stored leading many-body squared-singular-value levels to a maximum absolute error of $7.28e - 12$. The calculation reports the raw finite-size coefficients and their uncertainty without acceptance gates. The resulting central-charge coefficient is anomalous, while the ordered gap coefficients form a nearly equally spaced sequence.

I. PROTOCOL AND RECONSTRUCTION

The campaign uses maximally mixed initial conditions, hard support-truncated walls at $x = 5, 15$, $n_{\text{shell}} = 1$, $\alpha_1 = 1$, $\alpha_2 = 30$, raster- y measurements, perfect correction, and complex128 covariance evolution. Entropies and active-space occupations were saved every four cycles; the realized Born record weight $\omega(t)$ was saved every cycle. For each saved occupation spectrum $\{\nu_a\}$ we reconstruct the many-body levels

$$\ell_i = \log \sigma_i^2 = \log Z + \sum_a \log \max(\nu_a, 1 - \nu_a) - d_i, \quad (1)$$

where d_i are the lowest subset sums of $|\log[\nu_a/(1 - \nu_a)]|$. Exact zero and unit occupations retain infinite flip costs. We fit every trajectory separately in $W_1 = [N_y, 2N_y]$, $W_2 = [2N_y, 3N_y]$, and $W_3 = [3N_y, 4N_y]$. All confidence intervals resample complete trajectories, with 2000 replicates and seed 2026091702.

II. PURIFICATION AND TEMPORAL CONVERGENCE

The endpoint entropy is small at every circumference, while the full $S(t)/N_y$ curves provide a direct descriptive test of $z = 1$ scaling. Threshold-crossing times are linearly interpolated on the native four-cycle grid. We report the fitted threshold exponents and the paired $W_2 \rightarrow W_3$ shifts directly, without converting them into pass/fail decisions.

III. RECORD CLOSURE AND BOUNDARY LOCALIZATION

The leading reconstructed level is compared with $d\omega/dt$ in the same fit window. The endpoint ratio $-\omega(T)/T$ is retained only as a finite-depth diagnostic. The largest late-window relative mismatch between $d\omega/dt$ and λ_0 is $1.326e-03$. The first four soft modes are analyzed only through their aggregate subspace density and aggregate weight near both walls. This avoids nonphysical labels under rotations in nearly degenerate eigenspaces.

IV. FINITE-SIZE COEFFICIENTS

The late-window leading density is fit to

$$-\lambda_0/N_y = f_\infty + A_0/N_y^2, \quad (2)$$

with $1/N_y^4$ and successive $N_{y,\text{min}} = 24, 30, 36$ sensitivity fits. The primary result is $A_0 = 11.19$ with 95% interval $[-4.148, 27.04]$. This gives $\alpha c_{\text{eff}} = -21.374$ with 95% interval $[-51.646, 7.922]$. For the gaps we define $r_i \equiv x_i/c_{\text{eff}} = -A_i/(12A_0)$ and additionally quote $x_i/x_1 = A_i/A_1$:

i	A_i	αx_i	r_i	x_i/x_1
1	2.639	0.420	-0.0196	1.000
2	3.606	0.574	-0.0268	1.367
3	4.499	0.716	-0.0335	1.705
4	5.493	0.874	-0.0409	2.082

These are direct fit outputs, not gated claims. The negative r_i values result from the positive fitted A_0 and therefore expose the anomalous central-charge extrapolation. No conformal operator names are assigned because momentum, charge-sector, and two-wall symmetry labels were not saved.

V. RELATION TO EARLIER ENSEMBLES

The earlier $2N_y$ pilot and the independent three-size $4N_y$ purification campaign are compared only at the summary-ledger level. Their trajectories are never pooled with this primary seven-size campaign.

VI. CONCLUSION

The completed data validate the Gaussian many-body spectral reconstruction, deep-time purification, same-window record closure, and boundary localization. The ungated fits yield a negative αc_{eff} but clean positive ordered-gap coefficients. The accompanying CSV and JSON files retain the bootstrap intervals, temporal shifts, resolved fractions, and fit sensitivities needed to judge those estimates directly.

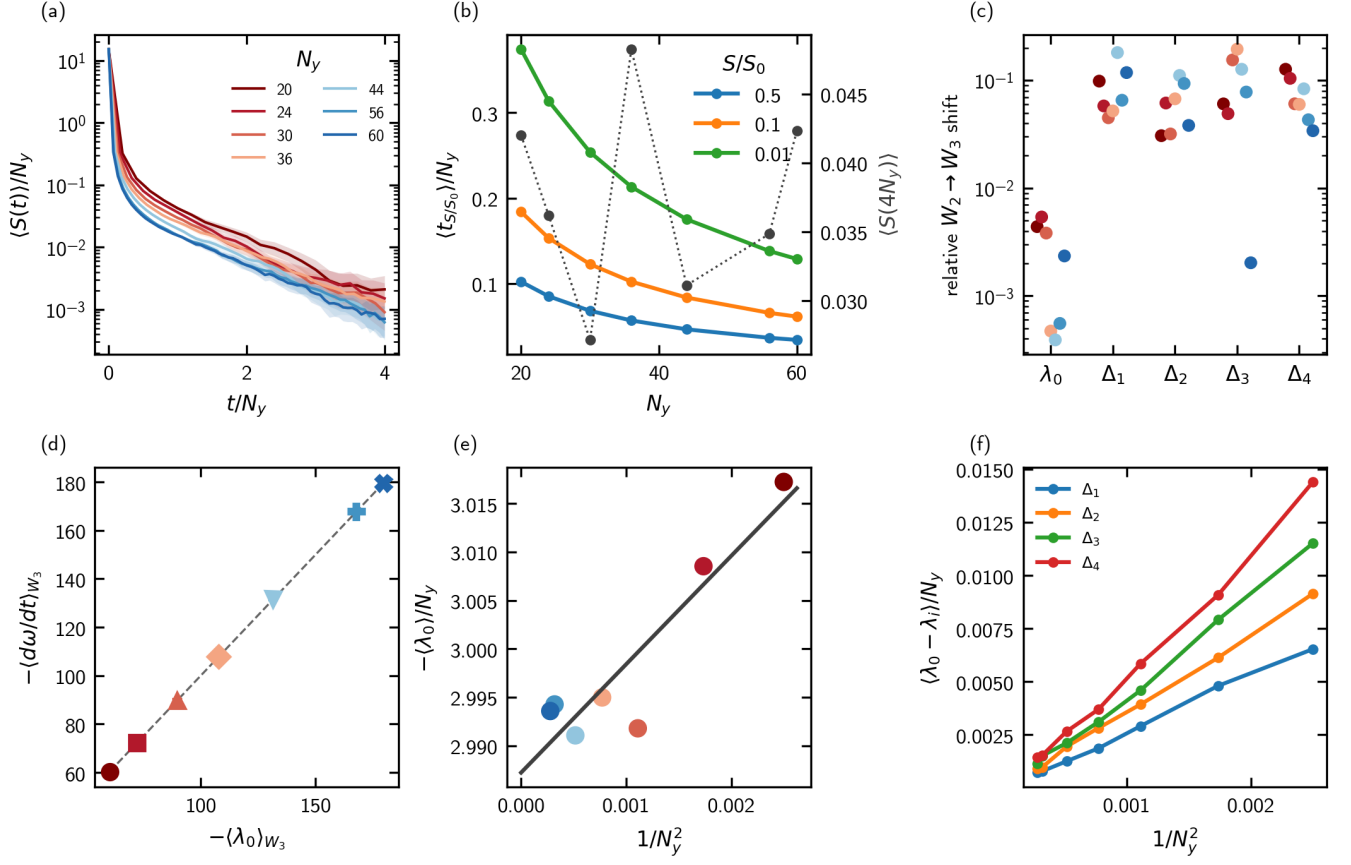


FIG. 1. Primary dynamical-critical results. (a) Entropy-density collapse with whole-trajectory bootstrap intervals. (b) Sample-wise purification times and endpoint residual entropy. (c) Relative $W_2 \rightarrow W_3$ shifts. (d) Same-window record-weight closure. (e) Leading finite-size fit. (f) Ordered-gap densities. Colors label circumference.

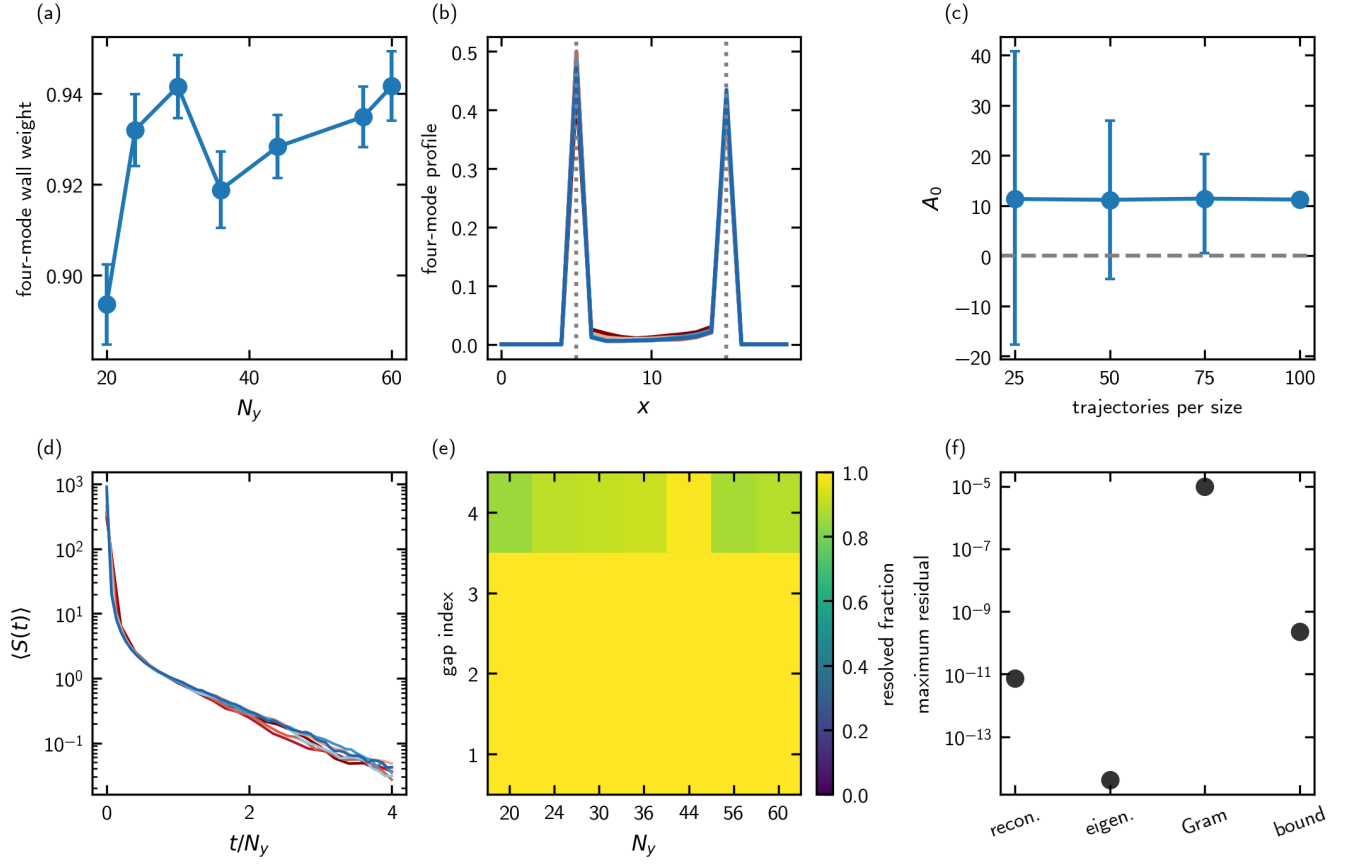


FIG. 2. Diagnostics. (a,b) Aggregate four-mode wall localization and transverse profile. (c) Random without-replacement sample convergence. (d) Unnormalized entropy dynamics. (e) Resolved fractions. (f) Maximum numerical residuals.